Jinsung Park

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Experience

Software Development Engineer Jul 2024 – Present
Amazon AWS (team: Data Warehouse as a Service)
- Automated account provisioning through state machine steps for Redshift, Bindle permission, Airflow, and DDB.
- Reconstructed the company's data collection system, utilizing unified authentication source GAM and GraphQL schema to expose application layer data.
- Maintained AWS revenue team's backend hexagonal architecture, streamlining access by migrating Isengard and Bindle permissions into the new Galaxi Portal.
- Drafted a 21-page Operational Reliance Review to support new feature rollouts, enabling the SUDS team to scale its low-latency applications with higher reliability.
- Monthly responsibility in maintaining up time across Airflow, Redshift, and Bedrock services on backend.

Research Assistant Nov 2021 – June 2024
UCSC Vertical, Architecture, Memories, and Algorithms (VAMA)
- Extended Python compiler for FIRRTL (HDL inter representation for RTL), adding 42 operations emitting C++ code.
- Produced a test bench to find 11+ bugs in a codebase implementing UInt and SInt types.
- Added 42 low-level critical operations into the Python library defined in FIRRTL to generate C++ code.
- Utilized Hypothesis and fuzzy algorithms to induce smart predictions in the model for better accuracy.

Projects

Wireless transcriber | *STM32, Embedded C, Zynq 7010, Verilog*
- Built a waveform generator with spare STM32 F410RB and a MCP4922 DAC to draw Bad apple on Analog oscilloscope. Found out the hard way that timing is heavily bottlenecked by the CPU, so I replaced the infra with Zybo-z7's FPGA fabric which allowed me to cap at 600KHz.
- Designed FIR module to reduce noise on top of CC1101 Driver to handle communication with over air.
- Optimized the timing of a custom software-defined waveform generator by utilizing a precisely tuned Timer Interrupt Service Routine (ISR) to stream data over SPI.

Chisel AI Accelerator | *Scala, Chisel, TPU, Verilog*
- Designed and implemented a DNN accelerator generator, achieving a processing speed of 5 TOPS during testing, enhancing performance metrics for real-time ML applications.
- Built and validated a Chisel-based hardware simulation pipeline for systolic arrays, implementing dot-product operations with clean, testable, and extensible code.

Optical Character Recognition for Trivia Games | *Java*
- HQ trivia and Cash Show were dumping cash to whoever could answer 12 questions successively, so I designed an optical character recognizing software for reading characters on screen and scraping websites.
- Parses trivia questions through OCR and provides answers based on web results.
- Correctly answers trivia questions with 3 choices within 5 seconds with 93% accuracy.

tryai.cinna.dev | *Rails, Ruby, PostgreSQL, Qdrant, Embeddings, RAG*
- Designed and implemented a complete Ruby on Rails full stack infrastructure from scratch using Figma and Excalidraw.
- Tool categorizes AI tools from tags and user suggestion. Also uses RAG on Qwen to filter top 3 entries from record.

Opensource Contributions

Xilinx/embeddedsw
- Contributed a bug fix to Xilinx open source embeddedsw repo to fix timer configuration in AXI timer so that initialization resets PWM and the two interface counters to prevent stale memory from corrupting new data. (PR #383)

Skills

Rust, Python, Verilog, Scala, C/C++, RISC-V, ESP32C3, FPGA, Xilinx, RTOS, Espressif, Vivado, STM32CubeIDE, Vitis, Kicad

Education

B.S. Computer Engineering + Economics (3.97 GPA) 2020 – 2024